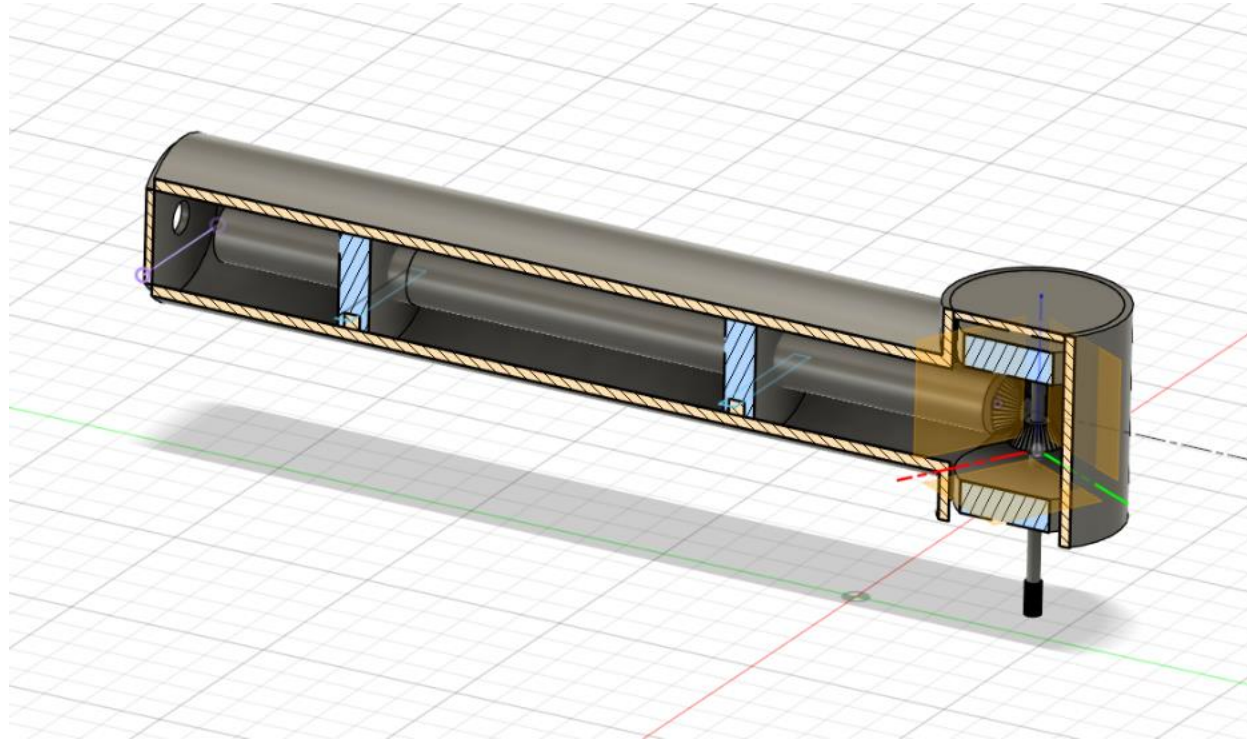


Design of a High-Efficiency Dental Rotary Cutting Tool

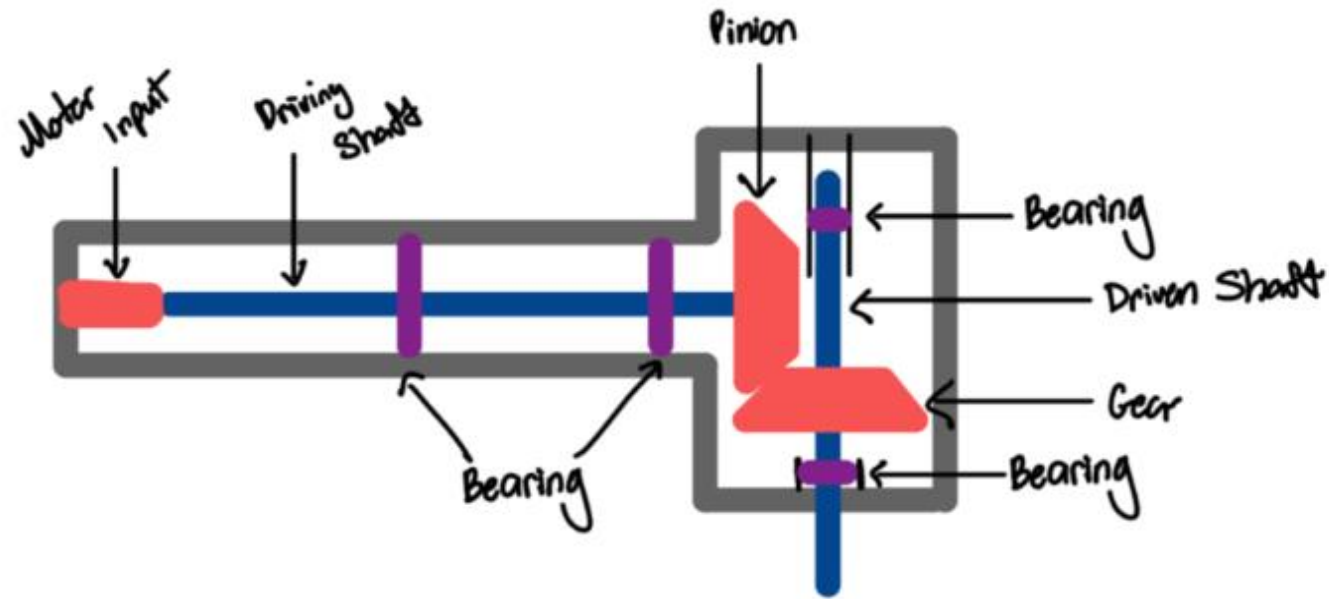
BE 3320

Brian Baudoin, Milain Marcel, Gavin Murray

Overall Design

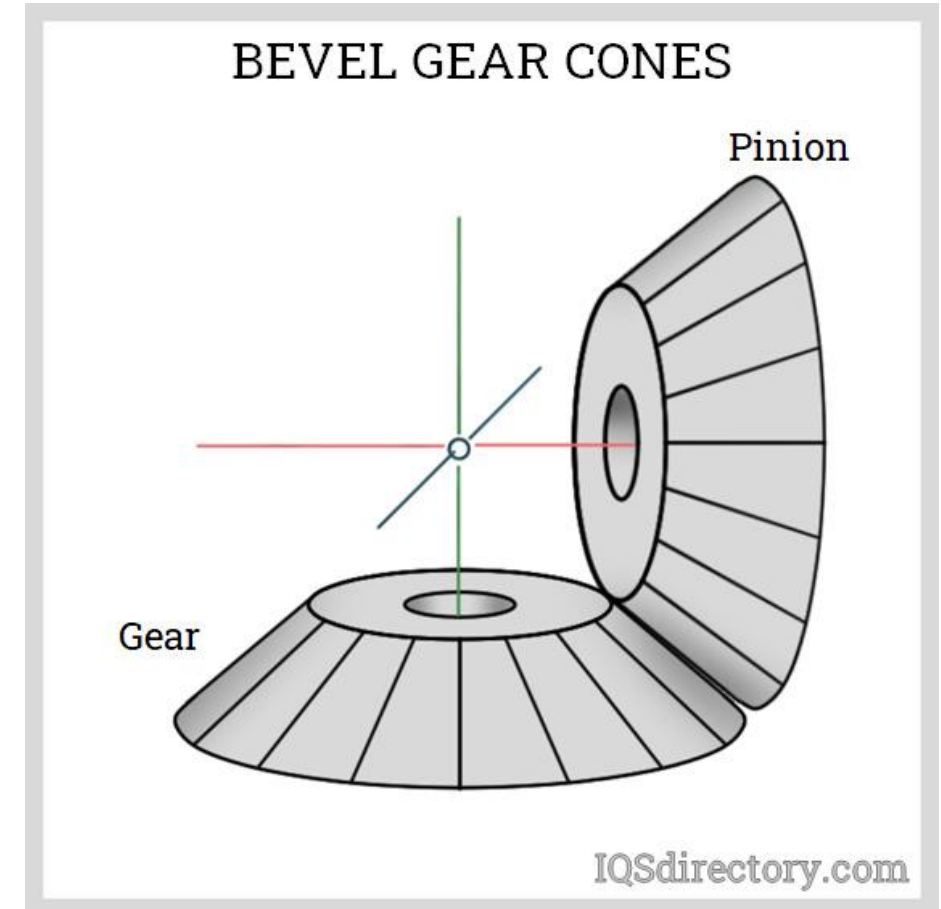


Powertrain Schematic



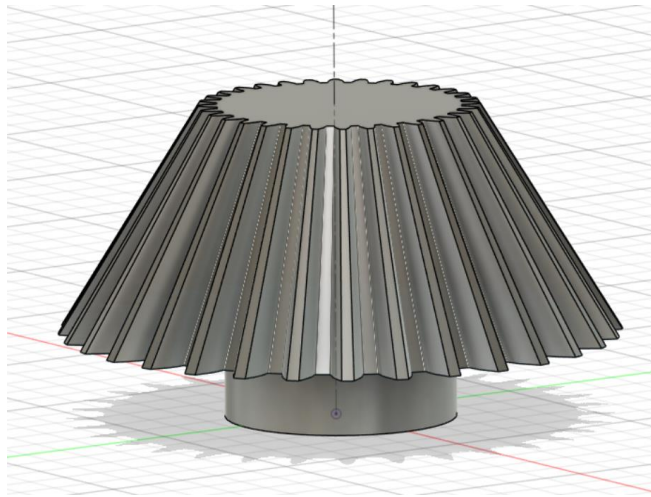
Powertrain Schematic

- Powertrain Configuration:
 - The system utilizes a precise mechanical gear train to transmit power from the motor to the cutting burr.
- Input source:
 - Electric Motor
- Transmission:
 - The driving shaft transmits torque longitudinally through the handpiece body
- Direction Change:
 - A 90 degree bevel gear set transfers rotation to the driven shaft



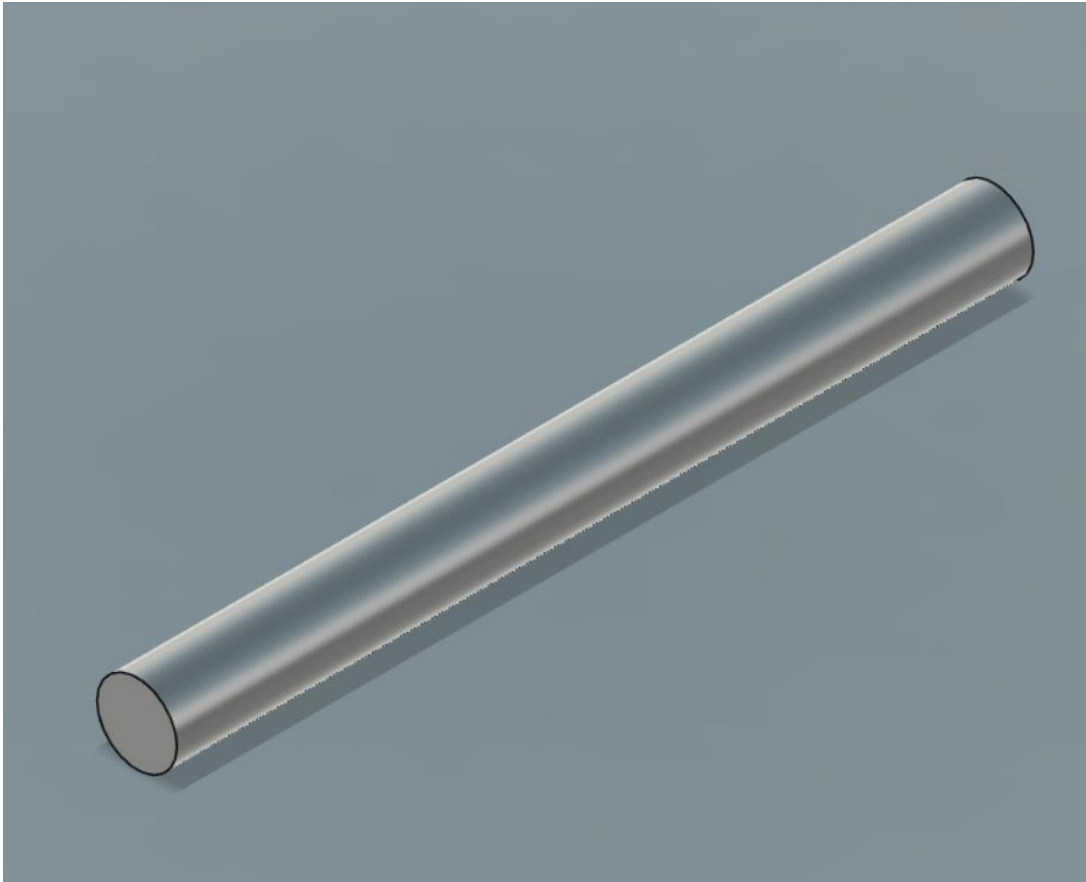
Gears and Gear Analysis

Needed Variables	Symbol	Value	units	Equation
Pitch line velocity	V	6283.185307	ft/min	$V = \pi \cdot d \cdot n / 12$
Transmitted Load	Wt	0.442536	lbf	$Wt = 33000 \cdot H / V$
Overload Factors	Ko	1		Table 15-2
Dynamic Factors	Kv	1.168076208		$Kv = (A + (\text{SQRT}(V)) / A)^B$
	A	92		Eq. 15-6, $A = 50 + 56(1-B)$
	B	0.25		Eq. 15-6, $B = 0.25(12 - Qv)^{(2/3)}$
Size factor	Ks	0.5		15-10, $Pd > 16$ teeth/in
Kmb- needed for Km	Kmb	1.1		one member straddle mounted
Load distribution factor	Km	1.100225		$Km = Kmb + 0.0036 \cdot F^2$
Lengthwise curvature factor	Kx	1		15-13
Geometry factor for Pinion	(J)P	0.215		Fig. 15-7
Geometry Factor for Gear	(J)G	0.215		Fig. 15-7
Elastic Coefficient	Cp	2290	psi ^{0.5}	15-21, assume steel
Geometry factor for pitting resistance	I	0.065		Fig. 15-6
Size factor for pitting resistance	Cs	0.46875		$Cs = 0.125F + 0.4375$
Crowning factor for pitting resistance	Cxc	1.5		15-12
Allowable bending stress number	Sat	13012	psi	$Sat = 44(HB) + 2100$ psi
Pinion Stress Cycle Factor for bending	(KL)P	0.9999634264		$1.683NL^{-0.0323}$
Gear Stress Cycle Factor for bending	(KL)G	0.9999634264		$1.683(NL/mG)^{-0.0323}$
Temperature Factor	KT	1		15-18, assume $t < 250$ degrees
Reliability Factor	KR	1		$0.50 - 0.25 \log(1-R)$
Allowable contact stress number	Sac	108188	psi	$Sac = 341(HB) + 23620$ psi
Pinion Stress-Cycle Factor for pitting	(CL)P	1.319634652		$3.4822NL^{-0.0602}$
Gear Stress-Cycle Factor for pitting	(CL)G	1.319634652		$3.4822(NL/mG)^{-0.0602}$
Hardness-Ratio Factor	CH	1		$CH = 1 + B1(mG - 1)$
	B1	0.00069		$B1 = 0.00898 (HBP/HBG) - 0.00829$
Contact Safety Factor	SH	2		
Reliability Factor	CR	1		$CR = \text{SQRT}(KR)$



Stress Analysis				
Variable	Symbol	Value	Units	Equation
Pinion bending				
Bending stress	(St)P	211.6181045	psi	$(St)P = (Wt/F) \cdot Pd \cdot Ko \cdot Kv \cdot ((Ks \cdot Km) / (Kx \cdot JP))$
Allowable bending stress	(Sw)P	6505.762052	psi	$(Sw)P = (Sat \cdot (KL)P) / (SF \cdot KT \cdot KR)$
Bending factor of safety for the pinion	(SF)P	7.841292876		$(SF)P = ((Sat \cdot (KL)P) / (KT \cdot KR \cdot (St)P))^{0.5}$
Gear bending				
Bending stress	(St)G	211.6181045	psi	$(St)G = (Wt/F) \cdot Pd \cdot Ko \cdot Kv \cdot ((Ks \cdot Km) / (Kx \cdot JG))$
Allowable bending stress	(Sw)G	6505.762052	psi	$(Sw)G = (Sat \cdot (KL)G) / (SF \cdot KT \cdot KR)$
Bending factor of safety for the gear	(SF)G	7.841292876		$(SF)G = ((Sat \cdot (KL)G) / (KT \cdot KR \cdot (St)G))^{0.5}$
Pinion wear				
Contact stress	(Sc)P	12700.78963	psi	$(Sc)P = Cp \cdot ((Wt/F \cdot dp^3) \cdot Ko \cdot Kv \cdot Km \cdot Cs \cdot Cxc)^{.5}$
Allowable contact stress	(Swc)P	71384.31688	psi	$(Swc)P = (Sac \cdot (CL)P \cdot CH) / (SH \cdot KT \cdot CR)$
Wear factor of safety for the pinion	(SH)P	3.352748991		$(SH)P = ((Sac \cdot (CL)P \cdot CH) / (KT \cdot CR \cdot (Sc)P))^{0.5}$
Gear wear				
Contact stress	(Sc)G	12700.78963	psi	$(Sc)G = Cp \cdot ((Wt/F \cdot dp^3) \cdot Ko \cdot Kv \cdot Km \cdot Cs \cdot Cxc)^{.5}$
Allowable contact stress	(Swc)G	71384.31688	psi	$(Swc)G = (Sac \cdot (CL)G \cdot CH) / (SH \cdot KT \cdot CR)$
Wear factor of safety for the gear	(SH)G	3.352748991		$(SH)G = ((Sac \cdot (CL)G \cdot CH) / (KT \cdot CR \cdot (Sc)G))^{0.5}$

Driving Shaft



Driven Shaft



Shaft Calculations

Finding Gear Forces

$$W_r = 0.442536 \text{ lbf}$$

$$\phi = 20^\circ$$

$$\Gamma = 45$$

$$W_r = W_t (\tan \phi) \cos \Gamma = 0.1139 \text{ lbf}$$

$$W_a = W_t (\tan \phi) \sin \Gamma = 0.1139 \text{ lbf}$$

Driving Shaft

$$F_1 = \sqrt{W_t^2 + W_r^2}$$

$$= \sqrt{0.44^2 + 0.1139^2}$$

$$= 0.457 \text{ lbf}$$

Driven Shaft

$$F_2 = \sqrt{W_r^2 + W_a^2}$$

$$= \sqrt{0.1139^2 + 0.1139^2}$$

$$= 0.161 \text{ lbf}$$

Max Max

Driving

$$\begin{aligned} M_{\max} &= F_1 (LAP) \\ &= -0.4569 \cdot (0.5) \\ &= -0.228 \text{ lbf} \cdot \text{in} \end{aligned}$$

$$M_a = 0.228$$

$$M_m = 0$$

$$T_a = 0$$

$$T_m = 0.01475$$

$$= 0.17716 \text{ lbf} \cdot \text{in}$$

Driven

$$\begin{aligned} M_{\max} &= \frac{F_2 (L_{ap})(L_{co})}{(L_{ap} + L_{co})} \\ &= \frac{-0.161 (1.964) (0.5)}{(0.5 + 0.5)} \\ &= -0.040267 \end{aligned}$$

$$M_a = 0.040267$$

$$M_m = 0$$

$$T_a = 0$$

$$T_m = 0.01475$$

$$= 0.17716 \text{ lbf} \cdot \text{in}$$

$$\sigma_a = \frac{\sigma_c (32) (M_o)}{\pi d_s^3}$$

$$\sigma_m = \frac{K_f (32) M_m}{\pi d_s^3}$$

$$\tau_a = \frac{K_{fs} (16) T_o}{\pi d_s^3}$$

$$\tau_m = \frac{K_{fs} (16) T_m}{\pi d_s^3}$$

B/C unnotched shafts

$$K_f = 1 \quad K_{fs} = 1$$

Shaft Diameters

$$d_{s1} = 2 \cdot \frac{1 \text{ in}}{25.4 \text{ mm}} = 0.2755 \text{ in}$$

$$d_{s2} = 1.6 \cdot \frac{1}{25.4} = 0.0629 \text{ in}$$

Shaft Calculations

Driving

$$\sigma_a = \frac{1(1)(32)(0.2285)}{\pi(0.2755)^3}$$

$$= 111.19 \text{ lbf} \cdot \text{in}^2$$

$$\sigma_m = \frac{1(32)(0)}{\pi(0.2755)^3}$$

$$= 0$$

$$\tau_a = \frac{1(16)(0)}{\pi(0.2755)^3}$$

$$= 0$$

$$\tau_m = \frac{1(16)(0.177)}{\pi(0.2755)^3}$$

$$= 43.07 \text{ lbf} \cdot \text{in}^2$$

Driven

$$\sigma_a = \frac{1(32)(0.0403)}{\pi(0.0629)^3}$$

$$= 1640.95 \text{ lbf} \cdot \text{in}^2$$

$$\sigma_m = \frac{1(32)(0)}{\pi(0.0629)^3}$$

$$= 0$$

$$\tau_a = \frac{1(16)(0)}{\pi(0.0629)^3}$$

$$= 0$$

$$\tau_m = \frac{1(16)(0.177)}{\pi(0.0629)^3}$$

$$= 3606.78 \text{ psi}$$

$$\sigma_a' = \sqrt{\sigma_a^2 + 3\tau_a^2}$$

$$\sigma_m' = \sqrt{\sigma_m^2 + 3\tau_m^2}$$

Driving shaft

$$\sigma_a' = \sqrt{(111.19)^2 + 3(0)^2}$$

$$= 111.19 \text{ psi}$$

$$\sigma_m' = \sqrt{0^2 + 3(43.07)^2}$$

$$= 74.60 \text{ psi}$$

Driven shaft

$$\sigma_a' = \sqrt{(1640.95)^2 + 3(0)^2}$$

$$= 1640.95 \text{ psi}$$

$$\sigma_m' = \sqrt{0^2 + 3(3606.78)^2}$$

$$= 6247.13 \text{ psi}$$

Material and Weight

Specifications

Gear Material	AISI 1095
Shaft Material	AISI 1095
Shell Material	stainless steel 316
Bearing Material	chrome steel
Burr Material	tungsten carbide
Bearing Type	No. 602, Miniature ball bearing

Mass Outputs					
Name	Symbol	Value	Unit	Conversion	Unit
Headpiece Mass	mhe	0.08014057	lb	36.3517618	g
Handpiece mass	mha	0.20583400	lb	93.3663025	g
Driving shaft mass	md	0.05002246	lb	22.6901864	g
Driven shaft mass	mdr	0.00160333	lb	0.7272720	g
Burr mass	mb	0.00043480	lb	0.1972263	g
Motor input shaft mass	mm	0.01667413	lb	7.5633850	g
Pinion mass	mp	0.03633957	lb	16.5	g
Gear mass	mg	0.04035353	lb	18.3	g
Bearing A mass	mba	0.00785848	lb	3.5646064	g
Bearing B mass	mbb	0.00785848	lb	3.5646064	g
Bearing C mass	mbc	0.00110229	lb	0.5	g
Bearing D mass	mbd	0.00110229	lb	0.5	g
Total mass	mtot	0.44932393	lb	203.8133350	g
power (hp) to weight(lbs) ratio	1:	5.332673754			

S_{ut} and S'_e

* Table A-20

AISI 1095 HR

$$S_{ut} = 120 \cdot 10^3 \text{ psi}$$

$$S_{ut} \leq 200 \cdot 10^3 \text{ psi}$$

$$\rightarrow S'_e = 0.5 S_{ut}$$

$$0.5(120 \cdot 10^3) = 60 \cdot 10^3 \text{ psi}$$

Correction Factors

$$K_a = a S_{ut}^b \quad S_{ut} = 120 \text{ kpsi}$$

$$\begin{aligned} \text{For HR} &\rightarrow a = 11 \text{ kpsi} \\ (\text{Table G-2}) \quad b &= -0.650 \end{aligned}$$

$$\begin{aligned} K_a &= 11 (120)^{-0.650} \\ &= 0.4890 \end{aligned}$$

$$K_b = 1 \quad (\text{assumed for unnotched shaft})$$

$$K_c = 1 \quad (\text{assumed for shaft in torsion and other loads})$$

$$K_d = 1 \quad (\text{assumed for } S_e \text{ from } S_{ut} \text{ at room temp.})$$

$$K_e = 0.868 \quad (\text{from Table G-4 for reliability of 95\%})$$

Solving for S_e with Marin's equation

$$S_e = K_a(K_b)(K_c)(K_d)(K_e)(S'_e)$$
$$= 0.4890(1)(1)(1)(0.868)(60)$$

$$S_e = 25.47 \text{ kpsi} = 25.47 \cdot 10^3 \text{ psi}$$

Verify DE Goodman Criteria

$$\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}} \geq \frac{1}{n} ; n \geq 2$$

* Driving Shaft

$$\frac{111.2}{(25.47 \cdot 10^3)} + \frac{74.6}{(120 \cdot 10^3)} \geq \frac{1}{2}$$

$$0.005 \geq 0.5$$

Shaft does not fail

* Driven Shaft

$$\frac{1640.95}{(25.47 \cdot 10^3)} + \frac{6247.13}{(120 \cdot 10^3)} \geq \frac{1}{2}$$

$$0.1165 \geq 0.5$$

Shaft does not fail

Actual Factor of Safety for Shafts

$$n = \frac{1}{\frac{\sigma'_a}{S_e} + \frac{\sigma'_m}{S_{ut}}}$$

Driving Shaft

$$\frac{1}{\frac{111.2}{(25.47 \cdot 10^3)} + \frac{74.6}{(120 \cdot 10^3)}} = n$$

$$n = 200.5$$

Driven Shaft

$$\frac{1}{\frac{1640.95}{(25.47 \cdot 10^3)} + \frac{6247.13}{(120 \cdot 10^3)}} = n$$

$$n = 8.6$$

Citations

IQS Directory. (n.d.). *Industrial Quick Search*. Types, Processes, and Applications for Bevel Gears.

<https://www.iqsdirectory.com/articles/gear/bevel-gear.html>

Thank You